

Logistic Regression – Solved Example 1

- The dataset of pass or fail in an exam of 5 students is given in the table.
 - Use logistic regression as classifier to answer the following questions.
- Calculate the probability of pass for the student who studied 33 hours.
 - At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

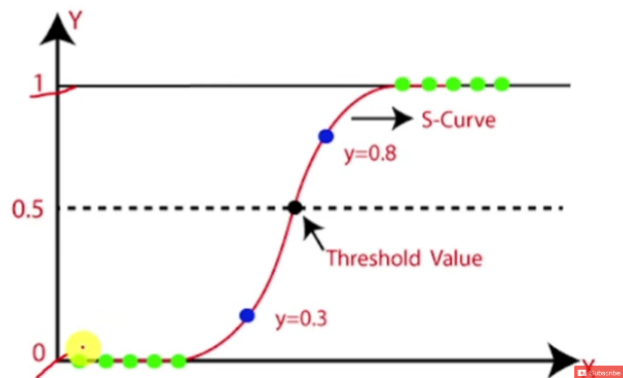
Assume the model suggested by the optimizer for odds of passing the course is,

$$\log(\text{odds}) = -64 + 2 * \text{hours}$$

Logistic Regression – Solved Example 1

- We use Sigmoid Function in logistic regression

$$s(x) = \frac{1}{1+e^{-x}}$$



Logistic Regression – Solved Example 1

- Calculate the probability of pass for the student who studied 33 hours.

$$p = \frac{1}{1+e^{-z}} \quad s(x) = \frac{1}{1+e^{-x}}$$

$$z = -64 + 2 * 33 = -64 + 66 = 2$$

$$p = \frac{1}{1+e^{-2}} = 0.88$$

- That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$\log(\text{odds}) = z = -64 + 2 * \text{hour}$$

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2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- $p = \frac{1}{1+e^{-z}} = 0.95$

- $0.95 * (1 + e^{-z}) = 1$

- $0.95 * e^{-z} = 1 - 0.95$

- $e^{-z} = \frac{0.05}{0.95} = 0.0526$

- $\ln(e^{-z}) = \ln(0.0526)$

$$\ln(e^x) = x$$

$$-z = \ln(0.0526) = -2.94$$

$$z = 2.94$$

Logistic Regression – Solved Example 1

- $z = 2.94$

- $\log(\text{odds}) = z = -64 + 2 * \text{hours}$

- $2.94 = -64 + 2 * \text{hours}$

- $2 * \text{hours} = 2.94 + 64$

- $2 * \text{hours} = 66.94$

- $\text{hours} = \frac{66.94}{2}$

- $\text{hours} = 33.47 \text{ Hours}$

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability